Shopping List App - Requirements specification

Version	Date of creation	Date last modified	Authors
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1. General information

1.1 Description of the system

The Shopping List App is a mobile application designed to support users in planning and managing their daily grocery shopping. The main business objective of the system is to reduce unnecessary expenses and food waste caused by impulsive purchases and poor planning. The application replaces traditional paper shopping lists with a digital solution that is always accessible on a mobile device.

The system allows users to create, edit, and manage shopping lists in a simple and intuitive way. By organizing products in one place, users can ensure they do not forget essential items while shopping. The application also supports better household stock management by helping users plan purchases in advance.

The system can be divided into the following functional modules:

- List Management Module - allows users to create, edit, and delete shopping lists and items.
- User Interface Module - provides a clear and user-friendly interface for quick access to lists during shopping.
- Data Storage Module - ensures that shopping lists are saved and accessible at any time.

Overall, the application focuses on simplicity, accessibility, and improving everyday shopping efficiency.

1.2 Target audience

The target audience of the Shopping List App includes individual consumers who regularly purchase groceries and household products. These are primarily people who want to better organize their shopping process, control their spending, and avoid forgetting important items.

The application is especially useful for:

- individuals managing their own household
- families responsible for weekly grocery shopping
- students and young professionals looking for simple organizational tools

The system is intended for everyday users rather than business clients, focusing on improving personal shopping habits and convenience.

1.3 Expected business benefits

The implementation of the Shopping List App is expected to bring several business-related benefits, primarily related to improving users' daily shopping experience and habits.

The key expected benefits include:

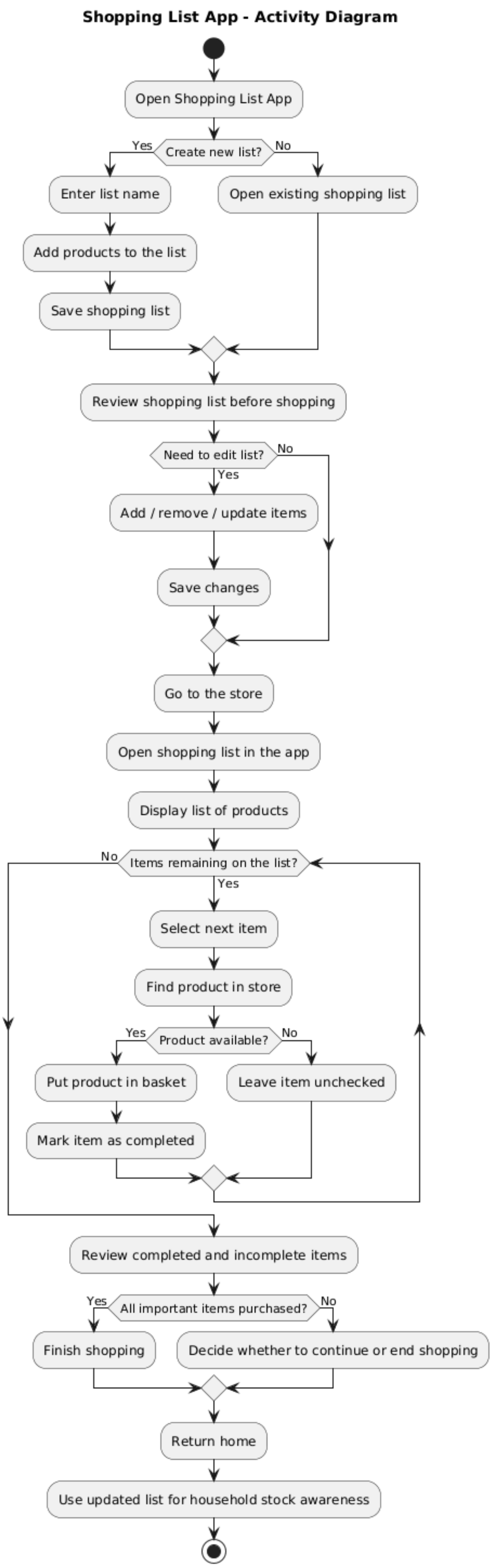
- Improved budget control - users can better plan their purchases and avoid unnecessary spending.
- Reduction of food waste - by buying only necessary items, users are less likely to waste unused products.
- Increased shopping efficiency - faster and more organized shopping process due to clear and accessible lists.
- Better household organization - improved management of products and supplies at home.

From a project perspective, the application also supports the development of practical software engineering skills and demonstrates the ability to deliver a complete digital product. These benefits help justify the implementation of the system, although their achievement is not a formal condition for system acceptance, which is based on meeting defined functional and non-functional requirements.

2. Business processes

BP01: Managing a shopping list

The user opens the Shopping List App on their mobile device. The user creates a new shopping list or selects an existing one. The system displays the list and allows the user to add, edit, or remove items. The system saves all changes and stores the list for future use. The user can continuously update the list based on current needs.


Shopping List App - Activity Diagram
Open Shopping List App
Create new list?
Yes
No
Enter list name
Open existing shopping list
Add products to the list
Save shopping list
Review shopping list before shopping
Need to edit list?
No
Yes
Add / remove / update items
Save changes
Go to the store
Open shopping list in the app
Display list of products
Items remaining on the list?
No
Yes
Select next item
Find product in store
Product available?
Yes
No
Put product in basket
Leave item unchecked
Mark item as completed
Review completed and incomplete items
All important items purchased?
Yes
No
Finish shopping
Decide whether to continue or end shopping
Return home
Use updated list for household stock awareness

3 Actors

ACT-001 User
A primary actor of the system who uses the application to create and manage shopping lists. The user interacts with the system to plan purchases, track items during shopping, and improve organization of household supplies.

Description of Responsibilities:

- Creating shopping lists: User creates new lists and adds products.
- Managing lists: User edits, updates, and deletes items.
- Using lists during shopping: User marks items as completed.
- Planning purchases: User prepares lists before shopping to avoid unnecessary spending.

Attributes:

- List name: Name assigned to the shopping list.
- List items: Products added to the list.
- Item status: Information whether the item is completed or not.

4 Business objects

OBJ-001 Shopping List
The Shopping List object represents a collection of products created by the user for a specific shopping purpose. It allows organizing and managing items that need to be purchased.

Relationships with Other Objects:

- User (ACT-001): Each Shopping List belongs to one User. The relationship is one-to-many, as one User can have multiple Shopping Lists.
- Item (OBJ-002): A Shopping List consists of multiple Items. The relationship is one-to-many.

Attributes:

- ID: Unique identifier of the shopping list.
- Name: Name of the list (e.g. “Weekly groceries”).
- Creation Date: Date when the list was created.

OBJ-002 Item
The Item object represents a single product entry within a shopping list. It contains information about a specific product that the user intends to purchase.

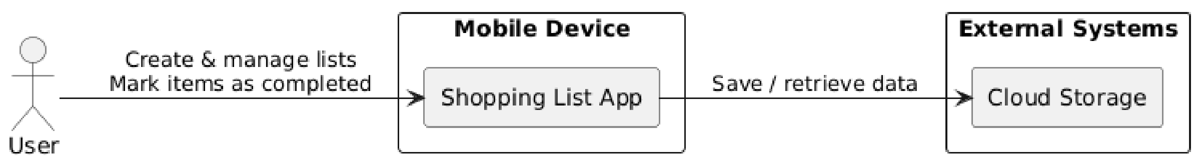
Relationships with Other Objects:

- Shopping List (OBJ-001): Each Item belongs to one Shopping List.

Attributes:

- ID: Unique identifier of the item.
- Name: Name of the product.
- Quantity: Number of units to purchase.
- Status: Indicates whether the item is completed (bought) or not.

5 Context diagram



6 Functional requirements

6.1 Requirements

US.01 - Create a shopping list

As a User, I want to create a new shopping list so that I can prepare products for a specific shopping trip.

DoD:

- User can create a new list.
- User provides a list name.
- List appears in the list overview after saving.
- Empty name triggers an error message.

US.02 - Name a shopping list

As a User, I want to enter a name for a shopping list so that I can easily identify it later.

US.03 - View shopping lists

As a User, I want to view all my shopping lists so that I can choose the correct one.

US.04 - Open shopping list

As a User, I want to open an existing shopping list so that I can review or modify it.

US.05 – Rename shopping list

As a User, I want to rename a shopping list so that I can keep list names clear and relevant.

US.06 – Delete shopping list

As a User, I want to delete a shopping list so that I can remove lists I no longer need.

US.07 - Add item to list

As a User, I want to add an item to a shopping list so that I can remember to buy it.

DoD:

- User can add an item to a selected list.
- Item must have a name.
- Item appears immediately after saving.
- Empty name triggers an error message.

US.08 - Edit item name

As a User, I want to edit an item name so that I can correct mistakes or update the product.

US.09 - Remove item from list

As a User, I want to remove an item from a shopping list so that I can keep the list up to date.

US.10 - Set item quantity

As a User, I want to specify the quantity of an item so that I know how many units to buy.

US.11 - Mark item as completed

As a User, I want to mark an item as completed so that I can track what I have already bought.

DoD:

- User can mark item as completed.
- Item status updates instantly.
- Completed item is visually different.
- Status is saved after reopening the app.

US.12 - Unmark item as completed

As a User, I want to unmark an item as completed so that I can correct mistakes during shopping.

US.13 - Distinguish item status

As a User, I want completed and not completed items to be visually distinguishable so that I can use the list more efficiently.

US.14 - View all items in list

As a User, I want to see all items assigned to a selected shopping list so that I can manage the full content of the list.

US.15 - Use list during shopping

As a User, I want to use my shopping list during shopping so that I can buy all necessary products.

US.16 - Real-time item updates

As a User, I want the system to update item status immediately after I mark it so that I always see the current state of the list.

US.17 - Identify missing items

As a User, I want to see which items are still missing so that I can finish shopping more effectively.

US.18 - Review shopping progress

As a User, I want to review completed and remaining items so that I have an overview of my shopping progress.

US.19 - Save shopping lists

As a User, I want the system to save my shopping lists so that I can access them later.

US.20 - Auto-save changes

As a User, I want the system to save all list changes automatically so that I do not lose my data.

US.21 - Persist lists after restart

As a User, I want previously created lists to remain available after reopening the application so that I can continue using them.

US.22 - Remove deleted data

As a User, I want deleted lists and items to be removed from storage so that outdated data is not displayed.

US.23 - Simple interface

As a User, I want the application interface to be simple and readable so that I can use it without difficulty.

US.24 - Clear UI actions

As a User, I want buttons and actions to be clearly described so that I can quickly understand how to use the application.

US.25 - Mobile usability

As a User, I want the shopping list to be easy to use on a mobile device so that I can operate it while shopping.

US.26 - Feedback messages

As a User, I want the application to display clear feedback messages after important actions so that I know whether an operation was successful.

UC1 - Managing a shopping list

Actors: User

Prerequisites:

- User has opened the application.
- At least one shopping list exists.

Main scenario

1. User selects a shopping list.
2. System displays list items.
3. User adds, edits, or removes an item.
4. User confirms the operation.
5. System saves changes.
6. System displays updated list.
7. Alternate scenarios (edge cases)

3a – Invalid or empty item name

3a.1 System displays error message.

3a.2 User corrects data.

3a.3 Return to step 4.

3b – Removing an item

3b.1 User selects the option to remove an item.

3b.2 System asks for confirmation.

3b.3 User confirms deletion.

3b.4 System removes item and updates list.

3b.5 Return to step 6.

3c – Cancel operation

3c.1 User starts add/edit action.

3c.2 User cancels the operation.

3c.3 System discards changes.

3c.4 System shows unchanged list.

3c.5 End of use case.

3d – No items in list

3d.1 User opens an empty shopping list.

3d.2 System displays message (e.g. “No items in the list”).

3d.3 User can add a new item.

3d.4 Return to step 3.

3e – Data save failure (e.g. crash / storage issue)

3e.1 System fails to save changes.

3e.2 System displays error message.

3e.3 User retries operation.

3e.4 Return to step 4.

3f – Duplicate item (optional behavior)

3f.1 User adds an item that already exists.

3f.2 System allows duplicate OR suggests merging quantity.

3f.3 System updates list accordingly.

3f.4 Return to step 6.

7. non-functional requirements

NFR1: Functional suitability

NFR for functional suitability has not been identified.

NFR2: Performance efficiency

The system shall display a shopping list and update item status in no more than 2 seconds under normal operating conditions, defined as using the application on a standard mobile device (e.g. iPhone with iOS 15 or higher) with up to 100 items per list and no background system overload.

Priority: High

NFR3: Compatibility

The system shall work correctly on iOS mobile devices running iOS version 15 or higher. The application shall be compatible with standard iPhone screen sizes (e.g. 4.7”–6.7”).

Priority: Medium

NFR4: Interaction capability

The user shall be able to perform key actions (creating a shopping list, adding an item, marking an item as completed, deleting an item) in no more than 3 steps.

Priority: High

NFR5: Reliability

After restarting the application, previously saved lists and item statuses shall still be available.

Priority: High

NFR6: Security

The system shall restrict access to user data by requiring authentication (username and password). User data shall not be accessible without proper authentication, and stored sensitive data shall be protected against unauthorized access.

Priority: Medium

NFR7: Maintainability

The system shall be developed in a modular way, separating list management, item management, data storage, and user interface logic. This shall allow future changes and fixes to be introduced without affecting the whole application.

Priority: Medium

NFR8: Flexibility

The system shall allow adding new features, such as item categories or reminders, without requiring changes to existing shopping lists and items. The application shall support extending functionality by adding new screens or options without affecting current features.

Priority: Low

NFR9: Safety

The system shall minimize the risk of accidental user errors by requiring confirmation before deleting a shopping list or important item. The application shall display clear messages before irreversible actions are performed.

Priority: Medium